

Three influential models, three representational designs

Geneformer

BERT-style encoder
rank-ordered gene tokens
no expression values
20,271 genes

scGPT

generative transformer
gene tokens + expression
value embeddings
60,694 genes

scFoundation

asymmetric encoder-decoder
continuous expression via
learned auto-discretisation
19,264 genes

identical four-metric geometry screen on the gene-embedding matrix

norm · centroid distance · cosine to centroid · isolation ($k = 10$); outlier if $|z| > 3$ on any metric

410 outliers

188 outliers

164 outliers

all three models

stable under
re-calling?

recur in ESM-2
sequence space?

associated with
ClinVar?

Geneformer only

costly to delete vs
matched controls?